

Senior Data & Platform Engineer and technical subject matter expert with nearly two decades of experience architecting distributed cloud pipelines, managing enterprise data integration, and ensuring high-stakes healthcare data integrity. Proven track record of navigating complex corporate acquisitions, establishing operational stability during crisis transitions, and developing custom backend tooling to bypass legacy bottlenecks and automate system recovery.

Work Experience

7/2020 – Present

Pittsburgh, PA

Senior Data Engineer

UPMC Enterprises

- **Architectural Migration & Cost Governance:** Spearheaded the migration of document ingestion workloads from AWS EKS to event-driven architectures, identifying and remediating high-frequency S3 polling inefficiencies to significantly reduce cloud API overhead.
- **Algorithmic Problem Solving:** Engineered the "Patient Identity Fixer," a specialized python utility that resolves "errored" patient records caused by erroneous HL7 ADT merge messages. Automated the re-linking of thousands of orphaned clinical documents, directly preventing data loss across disparate health systems.
- **Distributed ETL Orchestration:** Designed and maintained complex, multi-stage data pipelines utilizing AWS Step Functions, Lambda, and AWS Glue to process daily document rosters and health plan identifier CSVs. Engineered event-driven triggers via S3 notifications and SFTP landing zones.
- **Operational Tooling (Developer Experience):** Developed a custom Python diagnostic suite to aggregate and visualize execution statistics and error logs from distributed data jobs, standardizing incident response and reducing the cognitive load of manual system monitoring.
- **Technical Leadership & Mentorship:** Introduced advanced Object-Oriented Programming (OOP) principles, including custom class inheritance in Python, to streamline complex logic and reduce code duplication. Mentored junior engineering staff from initial onboarding to technical independence, accelerating their ability to autonomously design and deploy robust backend solutions.

7/2017 – 7/2020

Pittsburgh, PA

Integration and Data Services Manager

TeleTracking Technologies

- **Technical Scoping & Architecture:** Led technical architecture scoping for enterprise hospital implementations, rigorously evaluating client requests against core system capabilities to prevent scope creep and ensure accurate technical resource allocation.
- **Business Continuity & Crisis Management:** Assumed emergency management of a secondary technical team during a critical, overlapping absence of all senior personnel. Stabilized operations and ensured 100% business continuity for enterprise tools, earning explicit commendation from regional directors for exceptional crisis management.
- **Rapid Technical Onboarding & Mentorship:** Quickly acquired domain expertise in complex HL7 ADT message routing to serve as the primary subject matter expert and technical escalation point. Mentored newly onboarded engineering staff to maintain critical team workflows during severe resource shortages.

Skills

Languages & Frameworks

- Python
- Java
- SQL

Cloud & Infrastructure

- AWS Lambda, Step Functions, Glue, Fargate, EKS, S3, EventBridge, CloudFormation, CDK

Containerization & Deployment

- Docker
- Kubernetes (k3s, EKS)
- CI/CD Automation
- Git (GitLab/GitHub/Gitea)

Data Engineering & Architecture

- Event-Driven Architecture
- Distributed ETL Pipelines
- System Decoupling
- API Integration

Healthcare Domain Expertise

- HL7v2 (ADT Message Routing)
- FHIR
- HIPAA Compliance
- Enterprise Master Patient Index (EMPI) Integration

Interests

- Raspberry Pi/Home Lab Networking
- Programming Exercises
- SQL and Non-SQL Databases
- Science Fiction, Star Trek
- Olympic Lifting
- Photography
- Crocheting, Sewing

Matt Himrod (2)

- **Deployment & Infrastructure Support:** Directed emergency troubleshooting for the core interface engine during after-hours deployment windows, partnering directly with analysts and support desks to resolve critical integration blockers and finalize system upgrades.

11/2013 – 6/2017

Murrysville, PA

Integration & Deployment Manager

Philips (formerly medSage Technologies)

- **Technical Leadership & Corporate Scaling:** Promoted to manage the core integration team, successfully guiding data operations through a major corporate acquisition. Maintained strict operational continuity and data integrity while aggressively scaling deployment pipelines for new enterprise clients.
- **Enterprise Data Integration & Automation:** Architected and deployed automated data extraction pipelines across diverse client software environments. Engineered custom, automated reporting solutions on dedicated hardware to handle complex enterprise metric requirements.
- **Risk Management & Infrastructure Compliance:** Served as the primary technical Subject Matter Expert (SME) post-acquisition. Actively championed strict data compliance and aggressively advocated for the reinstatement of vital Disaster Recovery (DR) infrastructure to protect regulated healthcare data.

1/2008 – 11/2013

Murrysville, PA

Deployment Engineer

Philips (formerly medSage Technologies)

- **Tooling & Deployment Optimization:** Developed a foundational library of custom integration scripts (Perl) and complex SQL union queries functioning as self-populating validation checklists, dramatically reducing setup time and human error during client onboarding.
- **Pipeline Automation:** Engineered internal deployment tools to successfully bypass legacy API bottlenecks, automate client database queries, and streamline data ingestion to support an aggressive startup growth schedule.
- **End-to-End Implementation:** Directed end-to-end client onboarding across the country, serving as the technical bridge between external billing software feeds and internal patient outreach services.

Early Career Highlights

- **Case Western Reserve University (Teaching Assistant):** Held weekly recitation sessions and office hours, assigned homework and quizzes, answered questions, and proctored exams with a subset of students from the main class.
- **Case Western Reserve University (Systems Administrator):** Administered clinical software and research computing systems; established foundational expertise in HIPAA compliance, data security, and healthcare IT risk management.
- **Netronome Systems (Hardware QA Engineer):** Conducted hardware-level Quality Assurance for networking systems, utilizing low-level systems familiarity and C/C++.

Education

Case Western Reserve University

Cleveland, OH

Bachelor of Science in Engineering

- **Computer Engineering (major)**
- **Artificial Intelligence (minor)**